When Lenders Disagree

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Abstract

Banks that screen the same borrower with equal skill still reach different conclusions, because each runs its own credit check. This note builds the equilibrium of a credit market with that feature and solves it in closed form. The problem collapses onto two quantities — what a lender earns on a good borrower and on a bad one at a given rate — and a single index, the premium on a good borrower, then governs which lender quotes where. Exactly four configurations arise, and one explicit statistic decides whether a lender abandons good borrowers altogether. Where the two price ranges overlap, disagreement leaves no trace on the price of credit: it reallocates market shares, not prices. A second stage endogenises the credit check. Competition turns out to undo the incentive to screen wherever the two ranges share a rate, and we characterise exactly when the market collapses to no screening, and hence to no disagreement.

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Part I. The pricing game

1. The market

Banks and firms. A unit mass of firms borrows from a large number of risk-neutral banks. Each firm borrows one unit at rate r. A share π of firms are shoppers: they approach two banks and accept the lower offer. The other $1-\pi$ are captive: they approach one bank and accept its offer. A firm rejects any rate above its reservation rate. A bank therefore meets a firm either alone or against exactly one competitor, whom we call the rival. Since banks are many and each firm meets at most two, no single bank is large enough to move market aggregates — a fact we use only in Part II.

Types. A firm is good ($\theta = g$, share μ_g) or bad ($\theta = b$, share $\mu_b = 1-\mu_g$). Lending to it costs c_g or c_b; costs do not depend on the rate, as in the calibration. Good firms accept rates up to $\bar{r}^g$, which we call the **cap**, and bad firms up to $\bar{r}^b$, with

$$c_g < c_b < \bar{r}^g < \bar{r}^b.$$

Rates $r \le \bar{r}^g$ form the **pooling region**, where both types accept. Rates $\bar{r}^g < r \le \bar{r}^b$ form the **bad-only region**, where only bad firms accept.

Credit checks. Before quoting, each bank runs its own credit check on the firm in front of it and draws a signal $s \in \{g, b\}$, which we also call its verdict, with

$$\Pr(s = \theta \mid \theta) = q, \qquad q \in (\tfrac{1}{2}, 1].$$

The checks are run separately, so given the firm's type a bank's verdict and its rival's are independent draws. Precision is the same for both types and for every bank. Two banks of equal skill therefore disagree about the same firm with positive probability, and that is the feature this note is about.

One bank's signal. The shares of firms with each signal and the probability that the firm is good after each signal are

$$\sigma_g = q\mu_g + (1-q)\mu_b, \quad \sigma_b = 1-\sigma_g, \qquad \alpha^{s_g} = \frac{q\mu_g}{\sigma_g}, \quad \alpha^{s_b} = \frac{(1-q)\mu_g}{\sigma_b},$$

and the expected cost of lending after signal s is $c^{avg,s} = \alpha^s c_g + (1-\alpha^s)c_b$. The signal sorts firms but does not change how many good firms a bank meets in total:

$$\sigma_g\, \alpha^{s_g} + \sigma_b\, \alpha^{s_b} = \mu_g. \tag{1}$$

A bank and its rival. Averaging over the type gives the joint distribution:

	rival drew g	rival drew b
bank drew g	$q^2\mu_g + (1-q)^2\mu_b$	$q(1-q)$
bank drew b	$q(1-q)$	$(1-q)^2\mu_g + q^2\mu_b$

Two banks that meet the same firm disagree on a share $2q(1-q)$ of firms, whatever μ_g. For pricing, what matters is that given the firm's type the rival's signal does not depend on the bank's own: it is g with probability q for a good firm and $1-q$ for a bad firm.

Confident and doubtful lenders. We call a bank **confident** when it has just drawn a good signal on the firm at hand and **doubtful** when it has drawn a bad one, and we call α^s its confidence. These are informational states of one and the same bank, not two kinds of institution: every bank is confident about some firms and doubtful about others, and all banks are identical before the check.

Strategies and equilibrium. A bank that drew signal s quotes from a rate distribution F^s with no atoms: F^{s_g} after a good signal, F^{s_b} after a bad one. In a symmetric equilibrium every bank uses the same pair, and the profit $\Pi^s(r)$ after signal s satisfies

$$\Pi^s(r) = K^s \text{ on the support of } F^s, \qquad \Pi^s(r) \leq K^s \text{ at every } r.$$

Expected profit per firm examined is $\mathbb{E}[\Pi(q)] = \sigma_g K^{s_g} + \sigma_b K^{s_b}$.

Notation. A subscript g or b refers to the firm's type; a superscript s_g or s_b refers to the signal the bank drew.

symbol	meaning
α^s, $c^{avg,s}$	probability that the firm is good, and expected cost, after signal s
F^s, K^s	rate distribution, and equilibrium profit, after signal s
$Y_\theta(r)$	probability that the rival's offer to a type-θ firm is above r
$\Phi_\theta(r)$	profit on a type-θ firm quoted r
$\Delta(r)$	premium on a good firm, $\Phi_g(r) - \Phi_b(r)$
$A^s(r)$, $B^s(r)$	margin at stake on firms whose rival drew a good or a bad signal
φ^s	adverse selection after signal s
m	share of bad-signal quotes in the bad-only region
$\underline{r}$, $\underline{r}^b$	lowest rate in the market; lowest rate quoted in the bad-only region
ρ, ρ_g, ρ_-, r_2	rates that decide the configuration (Section 3.8)

2. Two kinds of firm, not two kinds of bank

2.1 The value of a good and of a bad firm

A shopper stays with a bank quoting r if the rival quotes more. The rival quotes from F^{s_g} or F^{s_b} depending on its own signal, and the distribution of that signal depends on the firm's type. The probability that the rival's offer to a type-θ firm is above r is

$$Y_g(r) = q\left[1 - F^{s_g}(r)\right] + (1-q)\left[1 - F^{s_b}(r)\right],$$
$$Y_b(r) = (1-q)\left[1 - F^{s_g}(r)\right] + q\left[1 - F^{s_b}(r)\right],$$

and the bank's profit on a type-θ firm quoted r is

$$\Phi_\theta(r) = \left[1 - \pi + \pi\, Y_\theta(r)\right](r - c_\theta). \tag{2}$$

Lemma 1 (profit). In the pooling region a bank that drew signal s earns

$$\Pi^s(r) = \alpha^s\, \Phi_g(r) + (1-\alpha^s)\, \Phi_b(r), \qquad r \le \bar{r}^g.$$

In the bad-only region good firms reject, and $\Pi^s(r) = (1-\alpha^s)\, \Phi_b(r)$.

Both signals face the same two functions and differ only in the weights. The lemma uses the independence of the signals given the type: once the type is known, the bank's own signal says nothing more about the rival's.

The same profit, counted by the rival's signal. A bank wins shoppers mostly when the rival drew the other signal, and those firms are worse than the bank's own signal suggests. The chance of winning and the margin therefore move together across types, so profit cannot be computed as an average chance of winning times an average margin. Counting firms by the rival's signal rather than by their own type gives the same total; Appendix B.2 counts it both ways.

2.2 Compact form

Substituting Y_g and Y_b into Lemma 1 gives, in the pooling region,

$$\Pi^s(r) = (r - c^{avg,s}) - \pi\left[A^s(r)\, F^{s_g}(r) + B^s(r)\, F^{s_b}(r)\right], \tag{3}$$

$$A^s(r) = \alpha^s\, q\,(r - c_g) + (1-\alpha^s)(1-q)\,(r - c_b),$$
$$B^s(r) = \alpha^s(1-q)\,(r - c_g) + (1-\alpha^s)\, q\,(r - c_b).$$

The bank starts from the profit it would earn if every firm accepted, $r - c^{avg,s}$. Good-signal quotes placed below r take away shoppers whose rival drew g; A^s is the margin at stake on those firms, and B^s is the same for rivals that drew b. Profit is linear in the two distributions. Two numbers recur below:

$$L = 1 - \pi q, \qquad M = 1 - \pi + \pi q,$$

the chances of keeping a good firm and a bad firm when every good-signal quote lies below r and every bad-signal quote above.

2.3 The two limits

- **No information, $q = \frac{1}{2}$.** Both signals carry the belief μ_g and $Y_g = Y_b$. Lemma 1 becomes the profit of a market with no information: $[1 - \pi + \pi(1 - F(r))](r - c^{avg})$ in the pooling region and $\mu_b[1 - \pi + \pi(1 - F(r))](r - c_b)$ in the bad-only region.
- **Perfect signal, $q = 1$.** $\alpha^{s_g} = 1$, $\alpha^{s_b} = 0$, $Y_g = 1 - F^{s_g}$ and $Y_b = 1 - F^{s_b}$. Each signal competes only with the same signal, and the model splits into two one-type markets, one per type.

Section 3.9 shows that the equilibria converge to these two benchmarks as well.

3. Equilibrium

Condition (A). Confident lenders do not prefer to lend to bad firms only:

$$K^{s_g} \;\geq\; (1 - \alpha^{s_g})(1 - \pi)(\bar{r}^b - c_b).$$

A sufficient condition, which can be checked before solving, is $\varphi^{s_g} \geq 0$, with φ^s defined in (6). Everything below assumes (A). When (A) fails, banks with either signal lend in the bad-only region; that case is not solved here (Section 3.9). Proof sketches are in Appendix A.

3.1 One index governs who quotes where

Subtracting the two profits in Lemma 1,

$$\Pi^{s_g}(r) - \Pi^{s_b}(r) = (\alpha^{s_g} - \alpha^{s_b})\,\Delta(r), \qquad \Delta(r) \equiv \Phi_g(r) - \Phi_b(r). \tag{4}$$

Δ is the **premium on a good firm**, and it is the only thing on which the two signals disagree. A confident lender prefers rates at which the premium is high, a doubtful lender

rates at which it is low. Write $\Delta^* \equiv (K^{s_g} - K^{s_b})/(\alpha^{s_g} - \alpha^{s_b})$.

Lemma 2 (sorting). Where F^{s_g} has mass, $\Delta \geq \Delta^*$. Where F^{s_b} has mass, $\Delta \leq \Delta^*$. Where both have mass, $\Delta = \Delta^*$, and Φ_g and Φ_b are each constant:

$$\Phi_g = \frac{(1-\alpha^{s_b})K^{s_g} - (1-\alpha^{s_g})K^{s_b}}{\alpha^{s_g} - \alpha^{s_b}}, \qquad \Phi_b = \frac{\alpha^{s_g}K^{s_b} - \alpha^{s_b}K^{s_g}}{\alpha^{s_g} - \alpha^{s_b}}.$$

The last clause does the work: the two signals can quote the same range of rates only if neither the value of a good firm nor the value of a bad firm changes across it.

3.2 How the premium moves

Lemma 3. With $G_1(r) = q(r-c_g) - (1-q)(r-c_b)$ and $G_2(r) = (1-q)(r-c_g) - q(r-c_b)$,

$$\Delta(r) = (c_b - c_g) - \pi\left[F^{s_g}(r)\,G_1(r) + F^{s_b}(r)\,G_2(r)\right].$$

$G_1 > 0$, so good-signal quotes placed below r always lower the premium at r. G_2 is positive below $r^\dagger = [qc_b - (1-q)c_g]/(2q-1)$ and negative above it, so bad-signal quotes lower the premium at low rates and raise it at high rates. At the lowest rate in the market nobody has quoted yet, and the premium takes its largest value, $c_b - c_g$.

3.3 Nobody leaves an idle stretch

Lemma 4. In the pooling region, the union of the two supports has no gap.

If nobody quoted on an interval (a, b) with quotes on both sides, a bank quoting a could move to just below b: it would face the same rival quotes and charge more.

3.4 Who opens the market and who closes it

Proposition 1. $\Delta^* < c_b - c_g$, and only confident lenders quote the lowest rate in the market.

Proposition 2. The highest rate quoted after a good signal is no higher than the highest rate quoted after a bad signal.

3.5 One statistic decides the top of the market: φ^{s_b}

At the highest rate in the market no rival quotes higher, so the bank earns only on captive firms.

Proposition 3. The highest rate in the market is $\bar{r}^g$ or $\bar{r}^b$, and

$$K^{s_b} = \begin{cases} (1-\pi)(\bar{r}^g - c^{avg,s_b}) & \text{if } \varphi^{s_b} \geq 0, \\ (1-\alpha^{s_b})(1-\pi)(\bar{r}^b - c_b) & \text{if } \varphi^{s_b} < 0, \end{cases} \tag{5}$$

where

$$\varphi^s \equiv (\bar{r}^g - c^{avg,s}) - (1-\alpha^s)(\bar{r}^b - c_b) = \alpha^s(\bar{r}^g - c_g) - (1-\alpha^s)(\bar{r}^b - \bar{r}^g). \tag{6}$$

In the second form c_b drops out: a bad firm is served in both regions, so its cost cancels. Quoting $\bar{r}^g$ keeps the good firms, with a margin $\bar{r}^g - c_g$ on each. Quoting $\bar{r}^b$ gives them up for the extra margin $\bar{r}^b - \bar{r}^g$ on bad firms. So $\varphi^s \geq 0$ exactly when the odds that the firm is good after signal s, $\alpha^s/(1-\alpha^s)$, are at least $(\bar{r}^b - \bar{r}^g)/(\bar{r}^g - c_g)$.

3.6 Who goes above the cap

Proposition 4. Under (A), only doubtful lenders quote in the bad-only region. The share of their quotes there is

$$m = \min\left\{1,\ \max\left\{0,\ \frac{-(1-\pi)\,\varphi^{s_b}}{\pi\, B^{s_b}(\bar{r}^g)}\right\}\right\}. \tag{7}$$

They quote on $[\underline{r}^b, \bar{r}^b]$, with

$$F^{s_b}(r) = 1 - \frac{1-\pi}{\pi q} \cdot \frac{\bar{r}^b - r}{r - c_b}, \qquad \underline{r}^b = \frac{(1-\pi)\,\bar{r}^b + \pi q\, m\, c_b}{1 - \pi + \pi q\, m},$$

and nobody quotes on $(\bar{r}^g, \underline{r}^b)$.

In the bad-only region every good-signal quote lies below, so a doubtful lender keeps a shopper only if the rival also drew b (probability q for a bad firm) and quotes higher. Profit falls at $\bar{r}^g$, where the good firms leave, and climbs back to K^{s_b} only at $\underline{r}^b$.

Three regimes, after a bad signal. Equation (7) sorts doubtful lenders exactly as adverse selection sorts a market with no information:

regime	condition	doubtful lenders lend in
full pooling, $m = 0$	$\varphi^{s_b} \geq 0$	the pooling region only
hybrid, $0 < m < 1$	$-\frac{\pi}{1-\pi} B^{s_b}(\bar{r}^g) < \varphi^{s_b} < 0$	both regions
bad-only, $m = 1$	$\varphi^{s_b} \leq -\frac{\pi}{1-\pi} B^{s_b}(\bar{r}^g)$	the bad-only region only

The threshold uses $B^{s_b}(\bar{r}^g)$, the margin on firms whose rival also drew a bad signal. If the rival always drew the same signal, as it does when the signal is public, $B^{s_b}(\bar{r}^g)$ would be $\bar{r}^g - c^{avg,s_b}$.

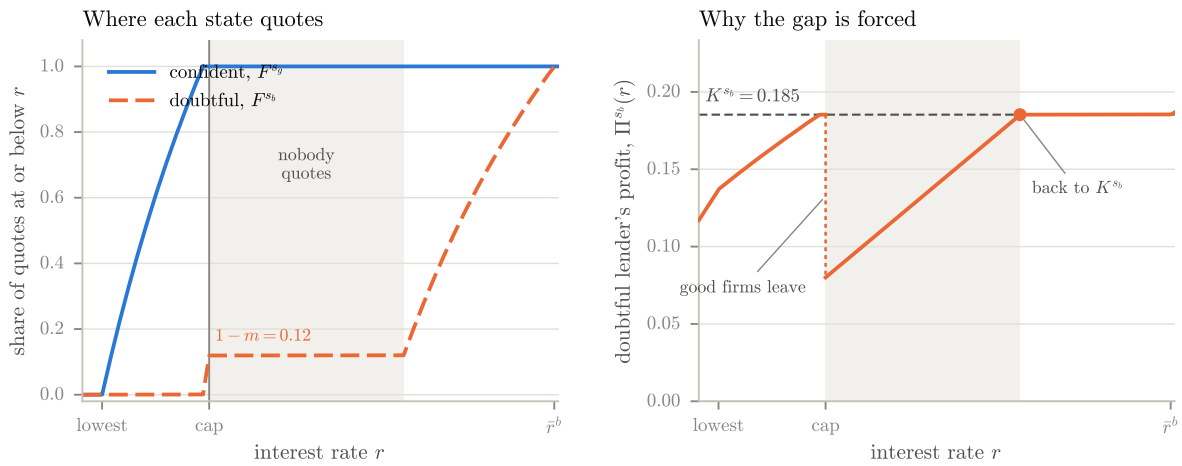


Figure 1: The retreat above the cap, for $\mu_g = 0.65$, $c_g = 0.05$, $c_b = 0.30$, $\pi = 0.40$, $\bar{r}^g = 0.45$, $\bar{r}^b = 0.80$ and $q = 0.75$, which give $m = 0.88$. Left: distribution functions of quoted rates. Doubtful lenders place 88% of their quotes above the cap $\bar{r}^g$; confident lenders quote only below it. In the grey band nobody quotes. Right: a doubtful lender's profit. It equals the equilibrium level K^{s_b} (dashed) at every rate it quotes, drops at the cap because the good firms leave, and climbs back to K^{s_b} only at $\underline{r}^b$. Quoting in between would earn less, so the gap is forced.

3.7 The contested stretch

By Lemma 2, Φ_g and Φ_b are constant where both signals quote. Their values are set at $\bar{r}^g$, where every good-signal quote lies below and a share m of bad-signal quotes lies above:

$$\Phi_g^{\top} = \Big[1 - \pi + \pi(1-q)\,m\Big]\,(\bar{r}^g - c_g), \qquad \Phi_b^{\top} = \Big[1 - \pi + \pi q\,m\Big]\,(\bar{r}^g - c_b).$$

Proposition 5. Where both signals quote,

$$\begin{aligned} F^{s_g}(r) &= \frac{1}{\pi}\Big[1 - \frac{1}{2q-1}\Big(\frac{q\,\Phi_g^{\top}}{r - c_g} - \frac{(1-q)\,\Phi_b^{\top}}{r - c_b}\Big)\Big], \\ F^{s_b}(r) &= \frac{1}{\pi}\Big[1 - \frac{1}{2q-1}\Big(\frac{q\,\Phi_b^{\top}}{r - c_b} - \frac{(1-q)\,\Phi_g^{\top}}{r - c_g}\Big)\Big], \end{aligned} \tag{8}$$

and $K^s = \alpha^s \Phi_g^{\top} + (1-\alpha^s)\Phi_b^{\top}$. Where confident lenders quote alone, below every bad-signal quote,

$$F^{s_g}(r) = \frac{(r - c^{avg,s_g}) - K^{s_g}}{\pi\, A^{s_g}(r)}, \qquad \underline{r} = c^{avg,s_g} + K^{s_g}. \tag{9}$$

When $m = 0$, each type of firm faces, on a shared range, the rate distribution of a one-type market with cost c_θ and top rate $\bar{r}^g$: $Y_\theta(r) = \frac{1-\pi}{\pi} \cdot \frac{\bar{r}^g - r}{r - c_\theta}$. The noise in the signals does not change the rates a type faces there; it changes which signal supplies them.

Proposition 6. A bank cannot withdraw quotes as the rate rises, so on a shared range both distributions must be non-decreasing. On (8), F^{s_b} always rises. F^{s_g} falls below

$$\rho_- = \frac{c_b - \tau c_g}{1-\tau}, \qquad \tau = \sqrt{\frac{(1-q)\,\Phi_b^\top}{q\,\Phi_g^\top}},$$

rises above it, and equals one at $\bar{r}^g$. A shared range therefore ends at $\bar{r}^g$ and starts no lower than ρ_-.

3.8 Putting it together: four configurations

Let $K^\circ = \alpha^{s_g}\Phi_g^\top + (1-\alpha^{s_g})\Phi_b^\top$, the profit of a confident lender at $\bar{r}^g$.

(iv) Bad-only, if $m = 1$. Doubtful lenders quote only in the bad-only region. Confident lenders quote alone on the whole range $[\underline{r}, \bar{r}^g]$, from (9) with $K^{s_g} = K^\circ$, and reach one exactly at $\bar{r}^g$.

If $m < 1$, two rates decide the configuration.

- ρ is where doubtful lenders would start quoting on the shared solution: $F^{s_b}(\rho) = 0$ in (8). It is the root in $(c_b, \bar{r}^g)$ of

$$q\left[(r-c_b) - \Phi_b^\top\right](r-c_g) = (1-q)\left[(r-c_g) - \Phi_g^\top\right](r-c_b).$$

- ρ_g is where confident lenders quoting alone at profit K° would have placed all their quotes: $F^{s_g}(\rho_g) = 1$ in (9), that is

$$\rho_g = \frac{K^\circ + \alpha^{s_g} L\, c_g + (1-\alpha^{s_g}) M\, c_b}{\alpha^{s_g} L + (1-\alpha^{s_g}) M}.$$

Proposition 7. With $m < 1$, the pooling region takes one of the following three configurations, and no other.

(i) Ranges meet, if $\rho_g < \rho$. Confident lenders would finish before doubtful lenders start, leaving rates nobody quotes, which Lemma 4 rules out. Instead K^{s_g} rises above K° until good-signal quotes end exactly where bad-signal quotes begin. At that rate $F^{s_g} = 1$ and $F^{s_b} = 0$, so

$$\rho = \frac{K^{s_b} + \alpha^{s_b} L\, c_g + (1-\alpha^{s_b}) M\, c_b}{\alpha^{s_b} L + (1-\alpha^{s_b}) M}, \qquad K^{s_g} = \alpha^{s_g} L\,(\rho - c_g) + (1-\alpha^{s_g}) M\,(\rho - c_b).$$

Confident lenders quote on $[\underline{r}, \rho]$ from (9). Doubtful lenders quote on $[\rho, \bar{r}^g]$ with

$$F^{s_b}(r) = \frac{\alpha^{s_b} L\,(r - c_g) + (1 - \alpha^{s_b}) M\,(r - c_b) - K^{s_b}}{\pi\, B^{s_b}(r)},$$

which reaches $1 - m$ at $\bar{r}^g$. The difference $K^{s_g} - K^\circ > 0$ is a rent: confident lenders hold a band of rates that doubtful lenders do not contest.

(ii) Ranges overlap, if $\rho_g \geq \rho$ and $\rho \geq \rho_-$. $K^{s_g} = K^\circ$. Confident lenders quote alone on $[\underline{r}, \rho]$ from (9), and both signals quote on $[\rho, \bar{r}^g]$ from (8).

(iii) Confident lenders pause, if $\rho_g \geq \rho$ and $\rho < \rho_-$. $K^{s_g} = K^\circ$. Sharing from ρ would require confident lenders to withdraw quotes. Instead F^{s_g} stays at $\bar{F} = F^{s_g}(\rho)$ until (8) climbs back to that level, at $r_2 \in (\rho_-, \bar{r}^g)$. On the pause $[\rho, r_2]$ only doubtful lenders quote,

$$F^{s_b}(r) = \frac{(r - c^{avg,s_b}) - K^{s_b} - \pi\,\bar{F}\,A^{s_b}(r)}{\pi\, B^{s_b}(r)},$$

and both signals quote on $[r_2, \bar{r}^g]$ from (8). The pause is a band of rates never quoted after a good signal, although lower and higher rates are.

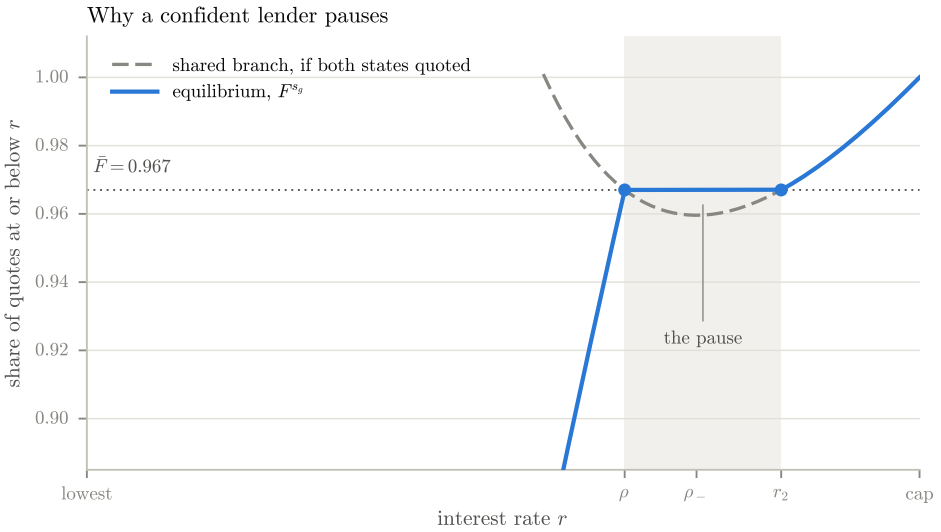


Figure 2: Why a confident lender pauses: the economy of Section 4 at $q = 0.80$, configuration (iii). The axis shows the share of a confident lender's quotes at or below each rate. Dashed grey: the shared branch (8), the distribution a confident lender would have to use if both states quoted from ρ on; it falls until ρ_- and climbs back only later. Solid blue: the equilibrium F^{s_g}. A distribution function cannot fall, so having met the shared branch at ρ, F^{s_g} holds the level $\bar{F}$ (dotted) and rejoins it at r_2. The dots mark the two crossings. In the grey band only doubtful lenders quote.

The equilibrium in two numbers. The whole solution is pinned by K^{s_b} and K^{s_g}. The statistic φ^{s_b} fixes K^{s_b} through (5)–(6); K^{s_b} fixes the share m placed above the cap through (7); m fixes the two values at the cap, $\Phi_g^\top$ and $\Phi_b^\top$, and with them K°. Then

either the solo branch (9) fits below ρ, in which case $K^{s_g} = K^\circ$ and the only question left is whether the two states share the range at once or after a pause, or it does not, in which case K^{s_g} rises above K° and the ranges merely meet. That gap, $K^{s_g} - K^\circ > 0$, is the rent disagreement pays a confident lender.

3.9 Two benchmarks: no information and a perfect signal

No information. At $q = \frac{1}{2}$ both signals carry the belief μ_g, one distribution serves both, and Lemma 1 gives the profit of a market with no information, so the equilibrium is the no-information one. As q falls to $\frac{1}{2}$ the equilibrium of this section converges to it: in the economy of Section 4 (full pooling at $q = \frac{1}{2}$), we find $K^{s_g} = 0.18752$ and $K^{s_b} = 0.18749$ at $q = 0.5001$, against $(1-\pi)(\bar{r}^g - c^{avg}) = 0.1875$ with no information. Near $q = \frac{1}{2}$, condition (A) depends on the regime that prevails at $q = \frac{1}{2}$. At $q = 0.52$ it holds in every random economy we draw with full pooling, in most with hybrid lending, and in none with bad-only lending. In those economies we do not solve the equilibrium.

Perfect signal. As $q \to 1$, $\alpha^{s_g} \to 1$, $\alpha^{s_b} \to 0$, and $Y_\theta \to 1 - F^{s_\theta}$: each signal competes only with itself, as in the equilibrium with a perfect signal. Confident lenders earn $K^{s_g} \to (1-\pi)(\bar{r}^g - c_g)$ and quote the good-type distribution of that equilibrium. After a bad signal, $\varphi^{s_b} \to \bar{r}^g - \bar{r}^b < 0$ and $B^{s_b}(\bar{r}^g) \to \bar{r}^g - c_b$, so

$$m \to \min\left\{1,\ \frac{(1-\pi)(\bar{r}^b - \bar{r}^g)}{\pi\,(\bar{r}^g - c_b)}\right\}, \qquad \underline{r}^b \to \max\left\{\bar{r}^g,\ c_b + (1-\pi)(\bar{r}^b - c_b)\right\}.$$

Doubtful lenders quote its bad-type distribution on $[c_b + (1-\pi)(\bar{r}^b - c_b), \bar{r}^b]$. When the lowest of these rates lies below $\bar{r}^g$, $m < 1$ in the limit: with no good firms left in their pool, doubtful lenders lose nothing at $\bar{r}^g$, and the band nobody quotes closes. So $m = 1$ in the limit only if $(1-\pi)(\bar{r}^b - \bar{r}^g) \geq \pi(\bar{r}^g - c_b)$.

4. Simulation

We simulate the equilibrium by solving Section 3 in the order set out in Appendix B.3. At every parameter value we report, we check it by brute force: on a fine grid of rates, profit from Lemma 1 equals K^s on each support and is no higher anywhere else.

Economy. $\mu_g = 0.65$, $c_g = 0.05$, $c_b = 0.30$, $\pi = 0.4$, $\bar{r}^g = 0.45$, $\bar{r}^b = 0.60$. Only the precision q varies. Condition (A) holds at every q in $[0.51, 0.99]$, and the brute-force check never finds an error above 10^{-15}.

4.1 Four equilibria

q	configuration	φ^{s_b}	m	K^{s_g}	K^{s_b}	$\mathbb{E}[\Pi(q)]$
0.70	(i) ranges meet	0.094	0	0.217	0.157	0.191
0.80	(iii) confident lenders pause	0.024	0	0.222	0.138	0.188
0.90	(ii) ranges overlap	−0.056	0.706	0.244	0.149	0.208
0.97	(iv) bad-only	−0.120	1	0.243	0.170	0.217

q	rates quoted after a good signal	rates quoted after a bad signal
0.70	$[0.314, 0.395]$	$[0.395, 0.45]$
0.80	$[0.302, 0.398] \cup [0.425, 0.45]$	$[0.398, 0.45]$
0.90	$[0.308, 0.45]$	$[0.433, 0.45] \cup [0.511, 0.60]$
0.97	$[0.297, 0.45]$	$[0.482, 0.60]$

- $q = 0.70$**, ranges meet.** Confident lenders quote on $[0.314, 0.395]$ and doubtful lenders on $[0.395, 0.45]$; the ranges touch at $\rho = 0.395$. After a bad signal the firm is still good with probability $\alpha^{s_b} = 0.44$, so $\varphi^{s_b} > 0$ and doubtful lenders pool: nobody quotes in the bad-only region.
- $q = 0.80$**, confident lenders pause.** Confident lenders quote up to 0.398, stop, and place their last 3.3% of quotes on $[0.425, 0.45]$. On the pause $[0.398, 0.425]$ only doubtful lenders quote; the density panel shows the gap in the blue curve.
- $q = 0.90$**, ranges overlap.** Confident lenders quote up to $\bar{r}^g$. Doubtful lenders are hybrid: they share $[0.433, 0.45]$ with confident lenders and place 71% of their quotes in the bad-only region, on $[0.511, 0.60]$. Nobody quotes on $(0.45, 0.511)$.
- $q = 0.97$**, bad-only.** Confident lenders quote alone in the pooling region, on $[0.297, 0.45]$. Doubtful lenders lend to bad firms only, on $[0.482, 0.60]$.

4.2 Checking the equilibrium at every rate

Profit equals K^s wherever the signal quotes and is lower everywhere else.

- **Good signal.** Below the lowest quote (0.302) profit rises with the rate. During the pause it dips below K^{s_g} by 0.0005 (0.2%): too little to see at full scale, but enough that no confident lender quotes there. In the bad-only region good firms reject, and profit falls to what bad firms alone pay.
- **Bad signal.** Profit rises until $\rho = 0.398$, stays at K^{s_b} up to $\bar{r}^g$, drops at $\bar{r}^g$ as the good firms in the pool reject, and climbs back only to 0.123 at $\bar{r}^b = 0.60$. That is why nobody quotes in the bad-only region at this q.

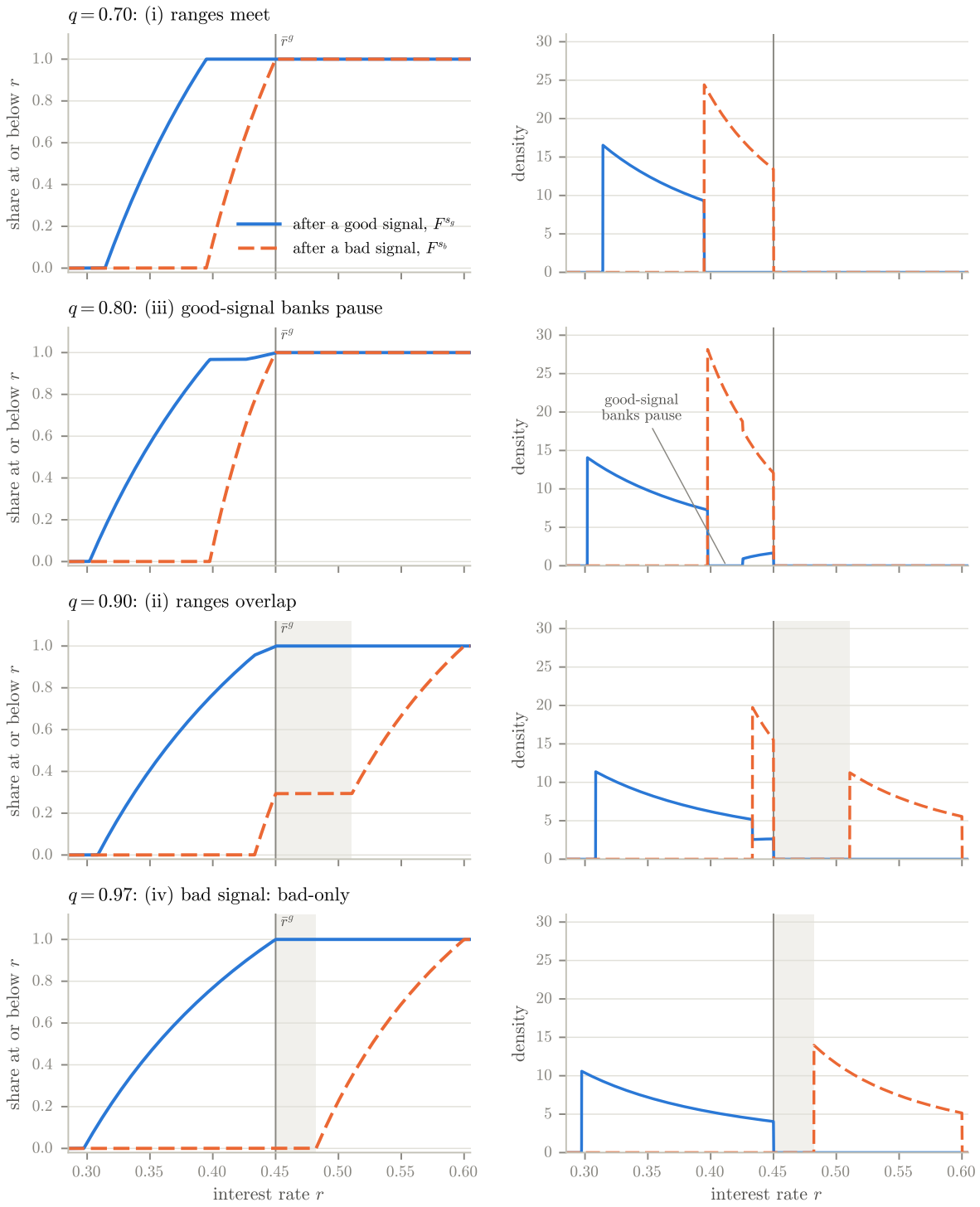


Figure 3: Equilibrium rate distributions at four values of q, one row each. Left: share of quotes at or below each rate. Right: density of quotes. Solid blue: after a good signal (F^{s_g}). Dashed orange: after a bad signal (F^{s_b}). Vertical line: $\bar{r}^g$, the top of the pooling region. Grey band: rates in the bad-only region that nobody quotes. Economy of Section 4.

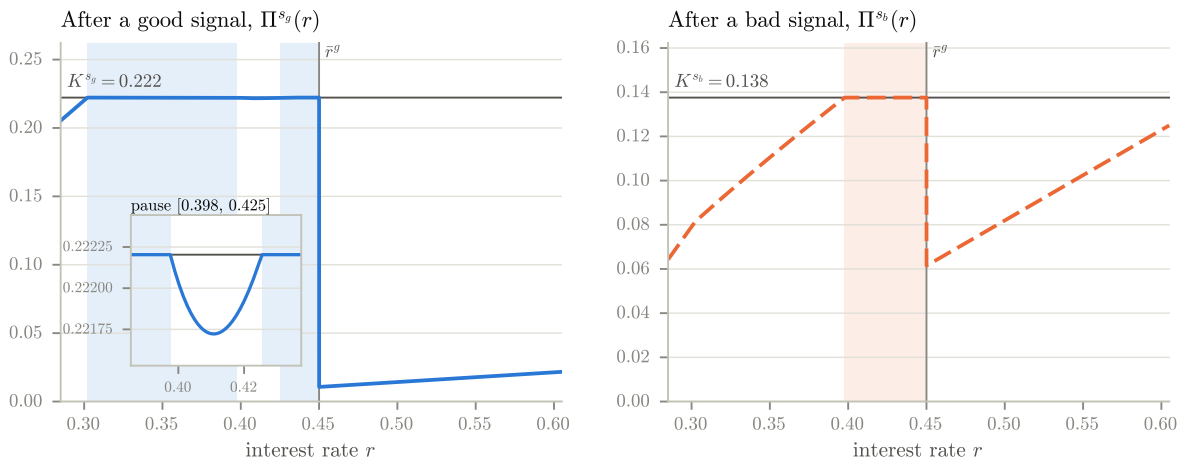


Figure 4: Profit after each signal at every rate, against the equilibrium distributions at $q = 0.80$ (the pause). Shaded: rates quoted after that signal. Horizontal line: equilibrium profit K^s. Vertical line: $\bar{r}^g$. Inset: the pause. Economy of Section 4.

4.3 The equilibrium as q rises

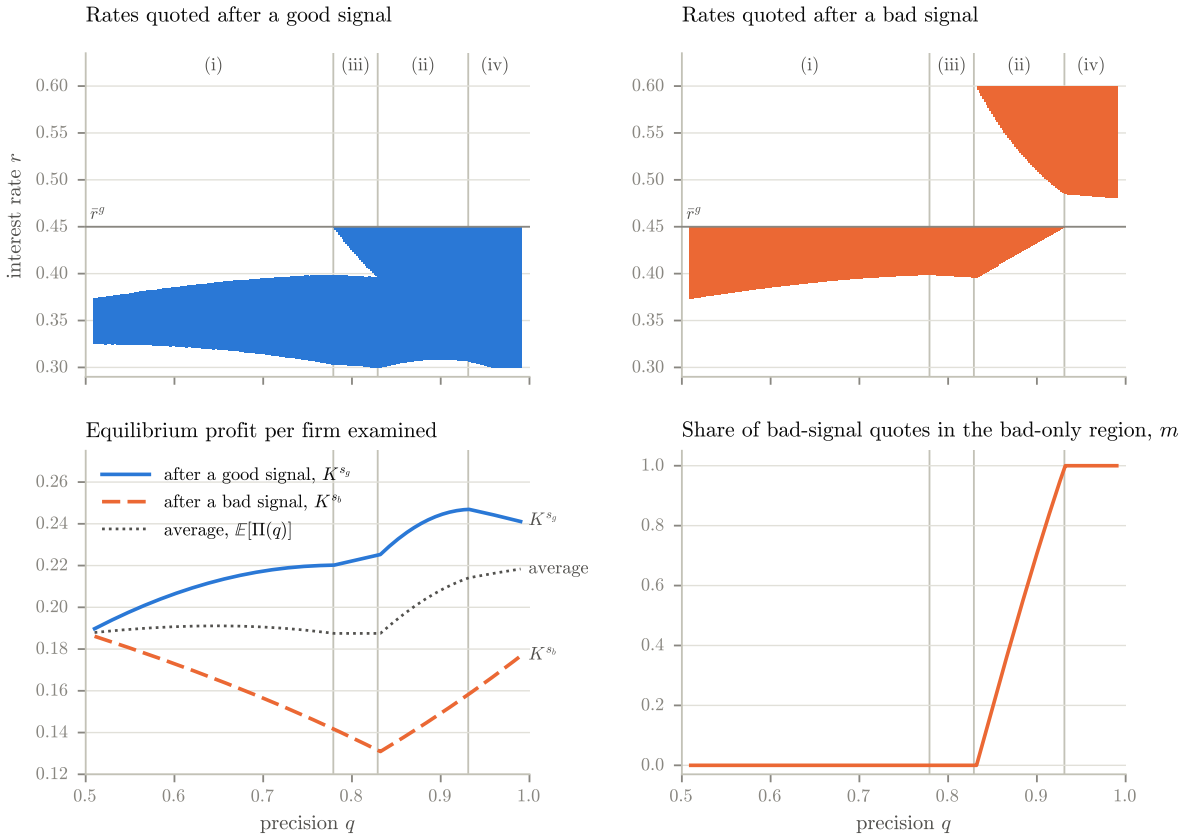


Figure 5: The equilibrium as q rises from 0.51 to 0.99, other parameters fixed. Top: rates quoted after a good signal (left) and after a bad signal (right), with the configuration above each range of q. Bottom left: equilibrium profit after each signal and expected profit. Bottom right: share of bad-signal quotes in the bad-only region. Vertical lines: changes of configuration, at $q = 0.78$, 0.83 and 0.93. Horizontal line: $\bar{r}^g$. Economy of Section 4.

- $q < 0.78$**: ranges meet.** Confident lenders quote the low rates and doubtful lenders the rates up to $\bar{r}^g$. The meeting rate rises from 0.373 to 0.399.
- $0.78 \leq q < 0.83$**: confident lenders pause.** The pause opens just below $\bar{r}^g$ ($[0.399, 0.449]$ at $q = 0.78$) and shrinks as q rises ($[0.397, 0.405]$ at $q = 0.82$) until it closes.
- $0.83 \leq q < 0.93$**: ranges overlap.** Confident lenders quote up to $\bar{r}^g$. From $q = 0.834$, $\varphi^{s_b} < 0$: doubtful lenders turn from full pooling to hybrid, m rises from 0 to 1, and their range in the pooling region shrinks toward $\bar{r}^g$.
- $q \geq 0.93$**: bad-only.** Doubtful lenders quote on roughly $[0.48, 0.60]$.

Profits:

- K^{s_g} rises from 0.190 at $q = 0.51$ to 0.247 at $q = 0.93$, then falls to 0.241.
- K^{s_b} falls from 0.186 to 0.131 at $q = 0.83$, while doubtful lenders still pool with fewer and fewer good firms. It then rises to 0.177 as they move to the bad-only region.
- Expected profit per firm examined stays between 0.187 and 0.191 up to $q = 0.83$, then rises to 0.218 at $q = 0.99$.

Part II. The information acquisition game

5. Information acquisition: profitable deviations

We ask whether a market's precision q^* is an equilibrium of the information-acquisition game: can a bank gain by choosing its own precision q_1 while its rival keeps q^* and the equilibrium rates $F^s(\cdot\,; q^*)$? With private signals the question has an exact answer.

5.1 One bank changes its credit check

What stays fixed. Banks are many and each firm meets at most two, so the deviating bank is of measure zero: its choice leaves the market's rate distributions where they were. Every other bank keeps precision q^* and quotes $F^s(\cdot\,; q^*)$, and the two verdicts on a firm are still independent given its type. Holding the rival's rates fixed is therefore no approximation at the level of the market; at the level of a single match it is the right test when the rival cannot see q_1 before it quotes.

What does not change. Given the firm's type, the rival's signal depends only on q^*. So Y_g, Y_b and the values Φ_g, Φ_b are those of the equilibrium at q^*, with $\Phi_g = 0$ in the bad-only region.

What changes. Only the deviator's own shares and beliefs, $\sigma_g(q_1)$, $\alpha^{s_g}(q_1)$ and $\alpha^{s_b}(q_1)$, from Section 1 with q_1 in place of q. After a signal that leaves belief α, the deviator quotes its best rate and earns

$$W(\alpha) = \max_r \ \Big[\alpha\, \Phi_g(r) + (1-\alpha)\, \Phi_b(r)\Big].$$

Its expected profit per firm examined is

$$\Pi^{dev}(q_1) = \sigma_g(q_1)\, W\Big(\alpha^{s_g}(q_1)\Big) + \sigma_b(q_1)\, W\Big(\alpha^{s_b}(q_1)\Big), \tag{10}$$

with $\Pi^{dev}(q^*) = \mathbb{E}[\Pi(q^*)]$. The case $q_1 = \frac{1}{2}$ is a bank that does not consult a signal. The participation decision, whether to pay C_s for the signal, compares $\Pi^{dev}(q^*) - C_s$ with $\Pi^{dev}(\frac{1}{2})$.

When the two precisions differ, a firm on which the banks disagree is no longer good with probability μ_g: the signal of the more precise bank counts more. At $q^* = 0.9$, a deviator that drew g while the rival drew b faces a good firm with probability 0.95 if $q_1 = 0.99$ and 0.45 if $q_1 = 0.80$. A bank wins shoppers mostly when the rival disagrees, so a sharper signal improves the firms it wins.

5.2 How deviation profit depends on precision

Let D^+ and D^- denote the slopes of Π^{dev} just above and just below q^*.

Proposition 8. Π^{dev} is convex and non-decreasing in q_1, and

$$D^{\pm} = \mu_g\Big[\Phi_g(r_g^{\pm}) - \Phi_g(r_b^{\pm})\Big] - \mu_b\Big[\Phi_b(r_g^{\pm}) - \Phi_b(r_b^{\pm})\Big]. \tag{11}$$

A little more precision makes the good signal more confident and the bad signal less. By (4), after a good signal the deviator moves to the rate with the highest premium in the good-signal range, $r_g^+ = \underline{r}$, and after a bad signal to the rate with the lowest premium in the bad-signal range, r_b^+. A little less precision does the opposite: r_g^- has the lowest premium in the good-signal range and r_b^- the highest in the bad-signal range. Two closed forms follow:

- If $m > 0$, every rate in the bad-only range has the lowest premium, with $\Phi_g = 0$ and $\Phi_b = (1-\pi)(\bar{r}^b - c_b)$:

$$D^{+} = \mu_g\,(\underline{r} - c_g) - \mu_b\Big[(\underline{r} - c_b) - (1-\pi)(\bar{r}^{b} - c_b)\Big].$$

- In configuration (iv), $r_g^{-} = \bar{r}^{g}$ and r_b^{-} is any rate in the bad-only range:

$$D^{-} = \mu_g\,(1 - \pi q^{*})(\bar{r}^{g} - c_g) - \mu_b\Big[(1 - \pi + \pi q^{*})(\bar{r}^{g} - c_b) - (1-\pi)(\bar{r}^{b} - c_b)\Big].$$

If $m = 0$, the lowest premium in the bad-signal range need not be at the top of the market. In configuration (iii) it lies inside the pause.

Proposition 9 (less precision). $\Pi^{dev}(q_1) = \Pi^{dev}(q^{*})$ for every $q_1 \le q^{*}$ if and only if some rate $r^{\sharp}$ is quoted after both signals in equilibrium, which is the case exactly when $m < 1$, in configurations (i) to (iii). The deviator then quotes $r^{\sharp}$ after both signals and earns $\mu_g\Phi_g(r^{\sharp}) + \mu_b\Phi_b(r^{\sharp})$, which by (1) does not depend on q_1. In configuration (iv), $\Pi^{dev}(q_1) < \Pi^{dev}(q^{*})$ for every $q_1 < q^{*}$, although Π^{dev} can be flat over a range of low q_1.

At a rate quoted after both signals, profit depends only on how many good firms the bank lends to, which is μ_g whatever its precision. Competition has already equalized the two signals' profits there, so the information is worth nothing. For the participation decision: whenever the two signals share a rate, a bank that stops paying for the signal earns what a bank that pays earns. No bank then pays any $C_s > 0$, and a market in which every bank consults the signal unravels.

5.3 Which markets can sustain screening

Let $C(q)$ be the cost of precision q, strictly increasing. The market's precision q^{*} survives if no bank gains by deviating once the cost is counted:

$$\Pi^{dev}(q_1) - C(q_1) \;\le\; \Pi^{dev}(q^{*}) - C(q^{*}) \qquad \text{for every } q_1.$$

Proposition 10. For $q^{*} > \frac{1}{2}$, a strictly increasing convex cost that sustains q^{*} exists if and only if $\Pi^{dev}(q_1) < \Pi^{dev}(q^{*})$ for every $q_1 < q^{*}$.

The construction is a cost that rises by S_{down} per unit of precision below q^{*} and by S_{up} above it, with

$$S_{down} = \frac{\Pi^{dev}(q^{*}) - \Pi^{dev}(\frac{1}{2})}{q^{*} - \frac{1}{2}}, \qquad S_{up} = \frac{\Pi^{dev}(1) - \Pi^{dev}(q^{*})}{1 - q^{*}}.$$

Because Π^{dev} is convex, S_{down} is the smallest loss per unit of precision dropped and S_{up} the largest gain per unit added, so $S_{up} \ge S_{down}$.

Proposition 11. Only two kinds of market sustain their precision against some strictly increasing convex cost: $q^* = \frac{1}{2}$, and configuration (iv), that is $\varphi^{s_b} \leq -\frac{\pi}{1-\pi} B^{s_b}(\bar{r}^g)$. Markets in which the ranges meet, overlap, or are separated by a pause are ruled out: a bank lowers its precision, quotes the shared rate, loses nothing, and saves cost.

In configuration (iv) no smooth cost works either: Π^{dev} has a corner at q^* with $D^- < D^+$, so a differentiable cost would need a marginal cost at q^* no larger than D^- and no smaller than D^+. The cost must rise slowly up to q^* and steeply beyond it, as with a screening technology that cannot go past q^*.

5.4 The two ends

Proposition 12 (starting from no information). At $q^* = \frac{1}{2}$, with $\mu_b = 1 - \mu_g$,

$$D^+(\tfrac{1}{2}) = 2\mu_g\mu_b \left[W'_+(\mu_g) - W'_-(\mu_g)\right] = \begin{cases} 2\pi\, \mu_g\mu_b\, (c_b - c_g) & \text{full pooling,} \\ 2\mu_g\mu_b \left[(c_b - c_g) + (1-\pi)(\bar{r}^b - c_b)\right] & \text{hybrid,} \\ 0 & \text{bad-only,} \end{cases}$$

where the regime is the one that prevails at $q = \frac{1}{2}$. The market with no information survives if and only if $C'(\frac{1}{2}) \geq D^+(\frac{1}{2})$. Under full pooling the gain from the first unit of precision is the product of three things: the share of shoppers (without them rates are not dispersed), $\mu_g\mu_b$ (there is nothing to learn about a pool of one type), and $c_b - c_g$ (what is at stake per firm). Under bad-only lending no good firm borrows, so there is nothing to sort and the first unit of precision is worth nothing.

Perfect signal. At $q^* = 1$, if $(1-\pi)(\bar{r}^b - \bar{r}^g) \geq \pi(\bar{r}^g - c_b)$, doubtful lenders lend in the bad-only region only (Section 3.9) and

$$D^-(1) = \mu_g\,(1-\pi)(\bar{r}^g - c_g) - \mu_b\Big[(\bar{r}^g - c_b) - (1-\pi)(\bar{r}^b - c_b)\Big].$$

A perfect signal survives if $C'(1^-) \leq D^-(1)$ (and the whole saving from any lower precision stays below the profit lost). If instead $(1-\pi)(\bar{r}^b - \bar{r}^g) < \pi(\bar{r}^g - c_b)$, the good-type and bad-type ranges share rates, Π^{dev} is flat on $[\frac{1}{2}, 1]$, and no increasing cost sustains a perfect signal.

In the economy of Section 4, $D^+(\frac{1}{2}) = 0.0455$ (full pooling) and $D^-(1) = 0.167$; a bank that drops from $q^* = 1$ to no information loses 4.8% of profit.

5.5 Simulation: deviations from four equilibria

The rival sits at each of the four equilibria of Section 4.1. Gains and losses are in % of $\Pi^{dev}(q^*)$, before the cost of precision.

q^*	configuration	$\Pi^{dev}(q^*)$	lower to $q_1 = 0.5$	raise to $q_1 = 0.999$	D^-	D^+
0.70	(i) ranges meet	0.191	0%	+23.1%	0	0.042
0.80	(iii) pause	0.188	0%	+20.8%	0	0.039
0.90	(ii) ranges overlap	0.208	0%	+10.8%	0	0.228
0.97	(iv) bad-only	0.217	−2.8%	+3.0%	0.170	0.225

- **Lowering precision.** In configurations (i) to (iii) profit does not change. The deviator quotes a shared rate after both signals: 0.395 at $q^* = 0.70$, 0.398 at $q^* = 0.80$, and 0.433 at $q^* = 0.90$. At $q^* = 0.97$ no rate is shared. For q_1 below 0.935 the deviator quotes $\bar{r}^g$ after both signals and loses 2.8%; between 0.935 and 0.97 it quotes $\bar{r}^g$ after a good signal and 0.482 after a bad one, and the loss shrinks to zero.
- **Raising precision.** After a good signal the deviator quotes the lowest rate in the market: 0.314, 0.302, 0.308 and 0.297. After a bad signal, at $q^* = 0.70$ it quotes $\bar{r}^g$, and at $q^* = 0.80$ it quotes 0.411, inside the pause (Proposition 8). In both markets it moves to $\bar{r}^b = 0.60$ from $q_1 = 0.83$ on, lending to bad firms only, which is the kink in the gain. At $q^* = 0.90$ and 0.97 it quotes in the bad-only range (0.511 and 0.482). The slopes in the table match (11).

5.6 Simulation: deviations at every market precision

- The gain from raising precision to 0.999 falls from 28.5% of profit at $q^* = 0.51$ to 7.3% at $q^* = 0.93$ and 0.9% at $q^* = 0.99$.
- The loss from lowering precision to 0.5 is zero up to $q^* = 0.930$ (configurations (i) to (iii)). It then rises to 4.1% at $q^* = 0.99$, and exceeds the gain from raising from $q^* = 0.972$ on.
- S_{up} stays between 0.11 and 0.23 at every q^*. S_{down} is zero up to $q^* = 0.930$ and stays below 0.02.
- In configuration (iv), $D^+ - D^-$ is at least 0.05 at every $q^* \geq 0.932$: a cost that sustains these markets must have a marginal cost that jumps at q^* by at least that much.

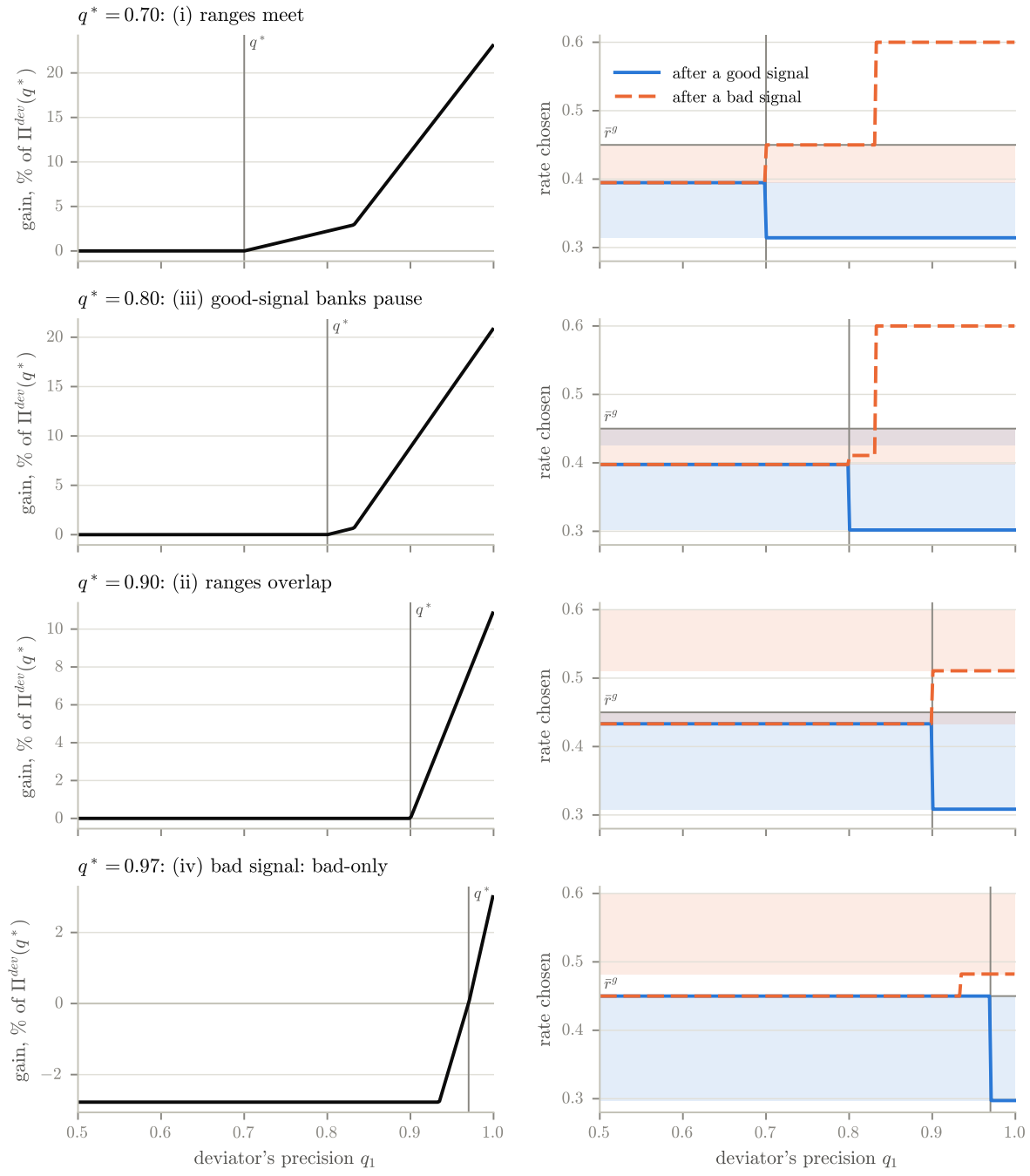


Figure 6: One bank deviates to precision q_1 while the rival stays at the equilibrium with precision q^* (vertical line), for the four equilibria of Section 4.1. Left: change in deviation profit, $\Pi^{dev}(q_1) - \Pi^{dev}(q^*)$, in % of $\Pi^{dev}(q^*)$, before the cost of precision. Right: rate the deviator quotes after a good signal (solid blue) and after a bad signal (dashed orange); when several rates tie, the lowest. Light bands: rates quoted in equilibrium after a good signal (blue) and after a bad signal (orange); where the bands overlap, both signals quote. Horizontal line: $\bar{r}^g$. Economy of Section 4.

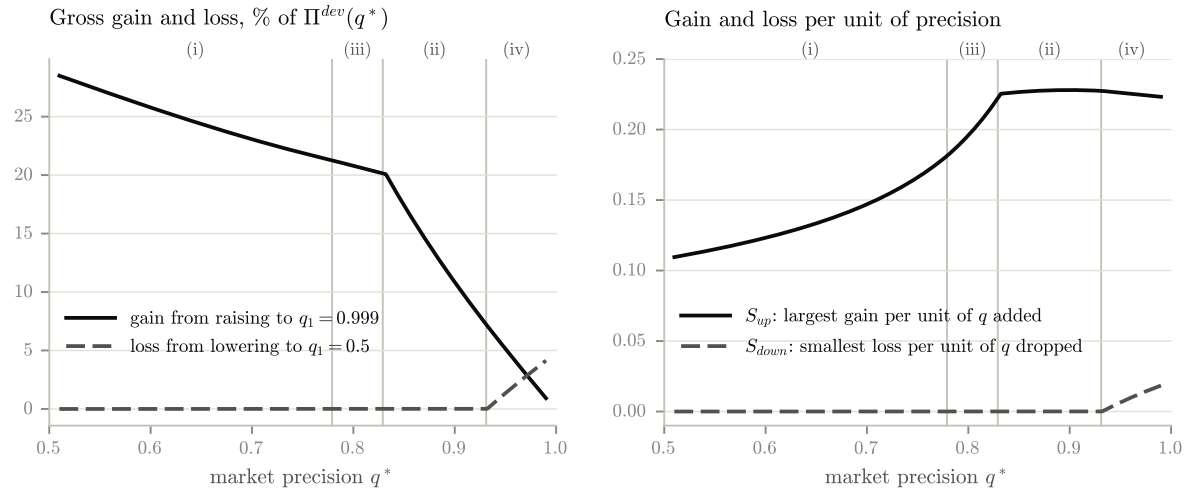


Figure 7: Gains and losses from a one-bank deviation, at every equilibrium precision q^* of the Section 4 economy. Left: gain from raising precision to $q_1 = 0.999$ and loss from lowering it to $q_1 = 0.5$, in % of $\Pi^{dev}(q^*)$, before the cost of precision. Right: S_{up} and S_{down}, with $q_1 = 0.999$ in place of 1. Vertical lines and labels: configurations, as in Figure 5.

6. Where this sits

The informational side of the model is Broecker's [3]: banks apply imperfect and independent credit tests and compete in rates, so that undercutting improves the average quality of one's own clientele, the feature that separates the problem from ordinary Bertrand competition. What we add is the demand side of the classic captive-and-shopper model of price dispersion, in the tradition of Varian [12], Burdett and Judd [5] and Narasimhan [11]. Neither literature on its own produces the structure above: Broecker's dispersion comes from the winner's curse with no search frictions, while the price-dispersion models have private values and no adverse selection. Endogenous screening is studied in a spatial setting by Hauswald and Marquez [7, 8]; Lester, Shourideh, Venkateswaran and Zetlin-Jones [9] share the ambition of joining search frictions to adverse selection, but with contract menus and private information on the borrower's side rather than noisy signals on the lender's.

The bridge comes from a third strand. Conditional on its verdict a bank is simply a seller with its own expected cost and its own captive base, so the pricing stage is a captive-and-shopper model with asymmetric sellers — studied by Baye, Kovenock and de Vries [2] and Golding and Slutsky [6], and recently in generality by Armstrong and Vickers [1] and Myatt and Ronayne [10], who find that asymmetric sellers compete over distinct price intervals rather than a common one, a pattern also visible in Bulow and Levin [4]. That is the same geometry as the configurations of Section 3.8, and the standard argument in that literature that no interval is left unquoted is exactly Lemma 4, which closes the construction. What does not carry over is that the value here is common rather than

private: the chance of winning depends on the firm's true type, which is what produces Lemma 1 and, through it, the premium and its law of motion.

Appendix A. Proof sketches

Lemma 1. Profit is the expectation over the firm's type of the chance of keeping the firm times the margin. Given the type, the rival's signal is g with probability q for a good firm and $1-q$ for a bad firm, whatever the bank drew; averaging the rival's distribution over its signal gives Y_θ. A captive firm accepts and a shopper accepts when not undercut, so the chance of keeping a type-θ firm is $1-\pi+\pi Y_\theta$ and the margin is $r-c_\theta$. The bank's own signal enters only through the weights α^s. In the bad-only region good firms reject, which deletes the first term.

Lemma 2. Equation (4) is Lemma 1 differenced. Where F^{s_g} has mass, $\Pi^{s_g} = K^{s_g}$ and $\Pi^{s_b} \leq K^{s_b}$, so $(\alpha^{s_g} - \alpha^{s_b})\Delta \geq K^{s_g} - K^{s_b}$. The same argument gives the second claim, and the two together give the third. Where both have mass, the two conditions $\alpha^s\Phi_g + (1-\alpha^s)\Phi_b = K^s$ are linear in (Φ_g, Φ_b) with determinant $\alpha^{s_g} - \alpha^{s_b} \neq 0$, so their solution is unique and does not depend on r.

Lemma 3. Substitute Y_g and Y_b into $\Delta = \Phi_g - \Phi_b$ and collect the terms in F^{s_g} and F^{s_b}. $G_1 > 0$ because $q > 1-q$ and $r - c_g > r - c_b$. G_2 is linear with slope $-(2q-1)$ and is zero at $r^\dagger$.

Lemma 4. Suppose nobody quotes on (a, b) in the pooling region while a is in some support. Moving those quotes from a to just below b leaves Y_g and Y_b unchanged. With Y_g, Y_b fixed, the derivative of Π^s in r is $\alpha^s[1-\pi+\pi Y_g] + (1-\alpha^s)[1-\pi+\pi Y_b] > 0$, so the move raises profit above K^s, which contradicts equilibrium.

Proposition 1. Let R_g be the top of the good-signal range, where $F^{s_g} = 1$. By Lemma 3, $\Delta(R_g) = (c_b - c_g) - \pi[G_1 + F^{s_b}G_2]$, and $G_1 + F^{s_b}G_2 \geq \min\{G_1, G_1 + G_2\} > 0$ because $G_1 + G_2 = c_b - c_g$. So $\Delta(R_g) < c_b - c_g$, and Lemma 2 gives $\Delta^* \leq \Delta(R_g) < c_b - c_g$. At the lowest rate both distributions are zero, so $\Delta = c_b - c_g > \Delta^*$: by Lemma 2 doubtful lenders cannot quote there.

Proposition 2. Suppose the good-signal top R_g exceeds the bad-signal top R_b. At R_g nobody quotes higher, so $\Delta(R_g) = (1-\pi)(c_b - c_g)$. At R_b, $F^{s_b} = 1$ and $F^{s_g} = 1-u$ with $u > 0$, so $\Delta(R_b) = (1-\pi)(c_b - c_g) + \pi u\, G_1(R_b) > \Delta(R_g)$. Both signals quote near R_b, so $\Delta(R_b) = \Delta^*$ and $\Delta(R_g) < \Delta^*$. But only confident lenders quote on $(R_b, R_g]$, where Lemma 2 requires $\Delta \geq \Delta^*$: a contradiction.

Proposition 3. Let R be the highest rate in the market; by Proposition 2 it is quoted after a bad signal, and nobody quotes above it. If $R < \bar{r}^g$, quoting slightly above R earns $(1-\pi)(R + \epsilon - c^{avg,s_b}) > K^{s_b}$; the same holds inside the bad-only region. So $R \in \{\bar{r}^g, \bar{r}^b\}$, and Lemma 1 with $Y_g = Y_b = 0$ gives (5). $R = \bar{r}^g$ requires that quoting $\bar{r}^b$ earn no more, $(1-\alpha^{s_b})(1-\pi)(\bar{r}^b - c_b) \leq (1-\pi)(\bar{r}^g - c^{avg,s_b})$, which is $\varphi^{s_b} \geq 0$. Expanding $c^{avg,s}$ gives the second form of (6).

Proposition 4. In the bad-only region the value of a bad firm is at most $(1-\pi)(\bar{r}^b - c_b)$: that is its value on the bad-signal range there, and it is lower on the band nobody quotes. A confident lender therefore earns at most $(1-\alpha^{s_g})(1-\pi)(\bar{r}^b - c_b)$ there, which (A) rules out. With every good-signal quote below, $Y_b = q(1-F^{s_b})$, and $(1-\alpha^{s_b})[1-\pi+\pi q(1-F^{s_b})](r-c_b) = K^{s_b}$ with K^{s_b} from (5) gives the distribution; $1-F^{s_b}(\underline{r}^b) = m$ gives $\underline{r}^b$. At $\bar{r}^g$, $F^{s_g} = 1$ and $F^{s_b} = 1-m$, so by (3), using $A^s + B^s = r - c^{avg,s}$, $\Pi^{s_b}(\bar{r}^g) = (1-\pi)(\bar{r}^g - c^{avg,s_b}) + \pi m B^{s_b}(\bar{r}^g)$. Setting this equal to $K^{s_b} = (1-\pi)(\bar{r}^g - c^{avg,s_b}) - (1-\pi)\varphi^{s_b}$ gives (7), bounded by $m \in [0,1]$. On $(\bar{r}^g, \underline{r}^b)$ a doubtful lender earns $(1-\alpha^{s_b})[1-\pi+\pi q m](r-c_b)$, which reaches K^{s_b} only at $\underline{r}^b$.

Proposition 5. By Lemma 2, Φ_g and Φ_b are constant on a range both signals quote. In each configuration of Proposition 7 that range ends at $\bar{r}^g$, where $F^{s_g} = 1$ and $F^{s_b} = 1-m$ give $Y_g = (1-q)m$ and $Y_b = qm$, so the constants are $\Phi_g^\top$ and $\Phi_b^\top$. Inverting (2) gives $1 - Y_\theta = [(r-c_\theta) - \Phi_\theta^\top]/[\pi(r-c_\theta)]$, and inverting the two equations for Y_g, Y_b gives (8). Where only confident lenders quote, $F^{s_b} = 0$ and (3) is linear in F^{s_g}, which gives (9); $F^{s_g} = 0$ gives $\underline{r}$.

Proposition 6. Differentiating (8), $\pi(2q-1)\,F^{s_g\prime} = q\Phi_g^\top/(r-c_g)^2 - (1-q)\Phi_b^\top/(r-c_b)^2$ and $\pi(2q-1)\,F^{s_b\prime} = q\Phi_b^\top/(r-c_b)^2 - (1-q)\Phi_g^\top/(r-c_g)^2$. So $F^{s_g\prime} \geq 0$ if and only if $[(r-c_b)/(r-c_g)]^2 \geq \tau^2$, that is $r \geq \rho_-$, because $(r-c_b)/(r-c_g)$ rises with r. $F^{s_b\prime} \geq 0$ if and only if $(\Phi_g^\top/\Phi_b^\top)[(r-c_b)/(r-c_g)]^2 \leq q/(1-q)$. The left side rises with r and at $\bar{r}^g$ equals $\frac{1-\pi+\pi(1-q)m}{1-\pi+\pi qm} \cdot \frac{\bar{r}^g - c_b}{\bar{r}^g - c_g} < 1 < \frac{q}{1-q}$, so this always holds. At $\bar{r}^g$, $F^{s_g} = 1$ by construction.

Proposition 7. By Proposition 1 the market opens with confident lenders alone, on (9). That phase ends when their quotes run out, at ρ_g, or when doubtful lenders become willing to quote. At the latter rate both indifference conditions hold with $F^{s_b} = 0$, so (9) and (8) agree there: it is ρ. If $\rho_g < \rho$, nobody would quote just above ρ_g, contradicting Lemma 4. Raising K^{s_g} lowers (9) at every rate until good-signal quotes end where bad-signal quotes begin, with $F^{s_g} = 1$ and $F^{s_b} = 0$; then $1-\pi+\pi Y_g = L$ and $1-\pi+\pi Y_b = M$, and $\Pi^{s_b}(\rho) = K^{s_b}$, $\Pi^{s_g}(\rho) = K^{s_g}$ give configuration (i). Above ρ, (3) with $F^{s_g} = 1$ is linear in F^{s_b}, and $(r - c^{avg,s_b}) - \pi A^{s_b}(r) = \alpha^{s_b} L(r-c_g) + (1-\alpha^{s_b})M(r-c_b)$. If $\rho_g \geq \rho$, the two branches meet at ρ with $F^{s_g} < 1$ and $K^{s_g} = K^\circ$. If $\rho \geq \rho_-$, F^{s_g} rises on (8) from ρ on: configuration (ii). If $\rho < \rho_-$, sharing would need F^{s_g} to fall, so confident lenders stop at $\bar{F}$ and only doubtful lenders quote, with (3) linear in F^{s_b}. Sharing resumes where (8) returns to $\bar{F}$; by Proposition 6, (8) falls until ρ_- and then rises to one, so it crosses $\bar{F} < 1$ once more, at $r_2 > \rho_-$. Above r_2, F^{s_g} rises, so there is no second pause: configuration (iii). If $m = 1$, no doubtful lender quotes in the pooling region, and (9) with $K^{s_g} = K^\circ$ reaches one at $\bar{r}^g$: configuration (iv).

Proposition 8. Since $\sigma_g \alpha^{s_g} = q_1\mu_g$, $\sigma_g(1-\alpha^{s_g}) = (1-q_1)\mu_b$, $\sigma_b\alpha^{s_b} = (1-q_1)\mu_g$ and $\sigma_b(1-\alpha^{s_b}) = q_1\mu_b$, quoting r_g after a good signal and r_b after a bad one earns $q_1\mu_g\Phi_g(r_g) + (1-q_1)\mu_b\Phi_b(r_g) + (1-q_1)\mu_g\Phi_g(r_b) + q_1\mu_b\Phi_b(r_b)$, which is linear in q_1. Π^{dev} is

the largest of these lines, so it is convex, and its one-sided slopes are the slopes of the lines through the best rates, (11). On the support of F^s, $\alpha\Phi_g + (1-\alpha)\Phi_b = K^s + (\alpha - \alpha^s)\Delta$, so for a belief just above α^s the best rate in the support maximizes Δ, and just below it minimizes Δ. Just above q^* the good-signal belief rises and the bad-signal belief falls, which gives r_g^+ and r_b^+; just below, the reverse. On the good-signal range Δ is largest at $\underline{r}$, where it equals $c_b - c_g$. For monotonicity, write $X = \mu_g[\Phi_g(r_g) - \Phi_g(r_b)]$ and $Z = \mu_b[\Phi_b(r_g) - \Phi_b(r_b)]$. Optimality of r_g after a good signal and of r_b after a bad one give $q_1 X + (1-q_1)Z \geq 0$ and $(1-q_1)X + q_1 Z \leq 0$. Subtracting, $(2q_1 - 1)(X - Z) \geq 0$, so the slope $X - Z$ is non-negative.

Proposition 9. W is the largest of functions linear in α, so it is convex. If $r^\sharp$ is best at both $\alpha^{s_b}(q^*)$ and $\alpha^{s_g}(q^*)$, it is best at every belief in between. Less precision moves both beliefs inside that interval, so the deviator quotes $r^\sharp$ after both signals and, by (1), earns $\mu_g\Phi_g(r^\sharp) + \mu_b\Phi_b(r^\sharp)$. If no rate is best at both ends, W is not linear on the interval, so it lies strictly below its chord inside the interval. $\Pi^{dev}(q_1)$ is an average of W at two interior beliefs with mean μ_g, which is below the chord at μ_g, that is below $\Pi^{dev}(q^*)$. If $m < 1$, the meeting rate (configuration (i)) or any rate on a shared range (configurations (ii) and (iii)) is quoted after both signals. If $m = 1$, the good-signal range ends at $\bar{r}^g$ and the bad-signal range starts at $\underline{r}^b > \bar{r}^g$, so no rate is shared.

Proposition 10. Necessity: if q^* is best and C is strictly increasing, then for $q_1 < q^*$, $\Pi^{dev}(q^*) - \Pi^{dev}(q_1) \geq C(q^*) - C(q_1) > 0$. Sufficiency: take C piecewise linear through q^* with slope S_{down} below and S_{up} above. Both slopes are positive, and convexity of Π^{dev} gives $S_{up} \geq S_{down}$, so C is strictly increasing and convex. Convexity also makes the chord slope $[\Pi^{dev}(q^*) - \Pi^{dev}(q_1)]/(q^* - q_1)$ rise with q_1, so it is at least S_{down}, and $C(q^*) - C(q_1) \leq \Pi^{dev}(q^*) - \Pi^{dev}(q_1)$. The argument above q^* is the same.

Proposition 11. At $q^* = \frac{1}{2}$ there is no lower precision, and a steep enough cost deters any increase. For $q^* > \frac{1}{2}$, Proposition 9 gives a flat Π^{dev} below q^* when $m < 1$, which violates Proposition 10, and $\Pi^{dev}(q_1) < \Pi^{dev}(q^*)$ below q^* when $m = 1$, which satisfies it. By (7), $m = 1$ is $\varphi^{s_b} \leq -\frac{\pi}{1-\pi} B^{s_b}(\bar{r}^g)$.

Proposition 12. At $q_1 = \frac{1}{2}$, $\sigma_g = \sigma_b = \frac{1}{2}$, both beliefs equal μ_g, and $d\alpha^{s_g}/dq_1 = -d\alpha^{s_b}/dq_1 = 4\mu_g\mu_b$. Differentiating (10), the terms in $W(\mu_g)$ cancel and $D^+(\frac{1}{2}) = 2\mu_g\mu_b[W'_+(\mu_g) - W'_-(\mu_g)]$. By the argument in Proposition 8, $W'_+(\mu_g)$ and $W'_-(\mu_g)$ are the largest and smallest premium over the rates quoted at $q = \frac{1}{2}$. With one distribution F, $\Delta = (1 - \pi F)(c_b - c_g)$ in the pooling region and $\Delta = -\Phi_b$ in the bad-only region. The largest premium is $c_b - c_g$, at the lowest rate. Under full pooling the smallest is $(1-\pi)(c_b - c_g)$, at $\bar{r}^g$. Under hybrid lending it is $-(1-\pi)(\bar{r}^b - c_b)$, anywhere in the bad-only range. Under bad-only lending every quoted rate has premium $-(1-\pi)(\bar{r}^b - c_b)$, so the difference is zero.

Perfect signal. At $q^* = 1$ in configuration (iv), apply the closed form for D^- in Proposition 8 with $q^* = 1$.

Appendix B. A numerical example

B.1 The economy

Appendices B.2 and B.3 use one economy: $\mu_g = 0.65$, $q = 0.75$, $c_g = 0.05$, $c_b = 0.30$, $\pi = 0.4$, $\bar{r}^g = 0.45$, $\bar{r}^b = 0.80$. The signal gives $\sigma_g = 0.575$, $\alpha^{s_g} = 0.848$ and $\alpha^{s_b} = 0.382$, and two banks that meet the same firm disagree on 37.5% of firms.

B.2 Profit counted by the rival's signal

Section 2.1 states that profit cannot be computed as an average chance of winning times an average margin. Take the 575 firms with a good signal out of 1,000 in this economy, at $r = 0.40$, when every good-signal quote lies below r and every bad-signal quote above. The bank keeps a firm with probability 0.6 if the rival drew g and with probability 1 if it drew b. The margin is 0.35 on a good firm and 0.10 on a bad one. Each cell is (firms) $\times$ (chance of keeping) $\times$ (margin):

	rival drew g	rival drew b	total
good firm	$365.6 \times 0.6 \times 0.35 = 76.8$	$121.9 \times 1 \times 0.35 = 42.7$	119.4
bad firm	$21.9 \times 0.6 \times 0.10 = 1.3$	$65.6 \times 1 \times 0.10 = 6.6$	7.9
total	78.1	49.2	127.3

- *Adding by rows* is Lemma 1. Within a row the margin is fixed and only the chance of keeping is averaged: 0.7 for good firms and 0.9 for bad firms, which are $1-\pi+\pi Y_g$ and $1-\pi+\pi Y_b$. Profit per firm is $0.848 \times 0.7 \times 0.35 + 0.152 \times 0.9 \times 0.10 = 0.221$.
- *Adding by columns* weights the rival's signal. The rival agrees with probability $387.5/575 = 0.674$, and then the average margin is 0.336. It disagrees with probability 0.326, and then the average margin is 0.2625. Profit per firm is $0.674 \times 0.6 \times 0.336 + 0.326 \times 1 \times 0.2625 = 0.221$.

B.3 Solving one equilibrium

The seven steps that solve the model, in the economy of B.1.

step	quantity	value
1	σ_g; α^{s_g}, α^{s_b}; c^{avg,s_g}, c^{avg,s_b}	0.575; 0.848, 0.382; 0.088, 0.204
2	φ^{s_b}; $K^{s_b} = (1-\alpha^{s_b})(1-\pi)(\bar{r}^b - c_b)$	-0.063; 0.185
3	$B^{s_b}(\bar{r}^g)$; m; $\underline{r}^b$	0.108; 0.881 (hybrid); 0.647
4	$\Phi_g^\top$, $\Phi_b^\top$; K°	0.275, 0.130; 0.2531
5	ρ; ρ_g (with $L = 0.7$, $M = 0.9$)	0.4441; $0.4433 < \rho$: configuration (i)
6	meeting rate ρ; K^{s_g}	0.4437; 0.2534
7	(A): $K^{s_g} \geq (1-\alpha^{s_g})(1-\pi)(\bar{r}^b - c_b) = 0.046$	holds; $\varphi^{s_g} = 0.286$

In this equilibrium:

- Confident lenders quote on $[0.341, 0.444]$.
- Doubtful lenders quote on $[0.444, 0.45]$, where F^{s_b} reaches $1 - m = 0.119$, and on $[0.647, 0.80]$. Nobody quotes on $(0.45, 0.647)$.
- Expected profit per firm examined is $\mathbb{E}[\Pi(q)] = 0.224$.

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